

Indian Monsoon & Climatology Essentials

ITC Z, jet streams, EL Niño links and monsoon mechanisms — yearly Prelims staple.

Portion focus ~44% · Topic: geo-physical

PYQ themes: Monsoon mechanism · Jet streams · El Niño/IOD · Cyclones

Why UPSC asks this

Monsoon integrates atmosphere dynamics with Indian livelihood — perfect for mechanism + statement questions.

Mechanism in exam language

Explain monsoon as seasonal reversal of winds driven by differential heating of land and ocean, modulated by the Intertropical Convergence Zone's northward shift, orography, and upper-air jets. Know onset/progress/withdrawal as map stories, not only definitions.

Teleconnections

El Niño–Southern Oscillation and Indian Ocean Dipole appear in recent papers. Treat correlations as probabilistic. Tropical cyclones in Bay of Bengal/Arabian Sea: conditions (warm SST, Coriolis, low shear) — link to disaster themes.

Pressure belts & world climate bridge

Hadley cells, horse latitudes, westerlies — still asked in physical geography blocks feeding monsoon understanding.

Map / list facts

- Western Ghats orographic rainfall
- Rain-shadow interiors
- Bay of Bengal cyclone nursery

Must remember

- Southwest monsoon June–Sep; retreat/NE monsoon affecting Tamil Nadu coast
- Tibetan heating + Mascarene High + Somali jet ingredients in standard models
- El Niño often correlated with weak monsoon — not iron law